

ERDENEMANDAL, Mongolia

WMO: 442370

Lat: 48.5294N

Lon: 101.3675E

Elev: 4951

StdP: 12.25

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.2	-21.0	-30.5	1.2	-23.1	-26.9	1.5	-18.9	23.4	10.0	20.8	10.0	4.0	230	0.681

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.6	81.9	62.4	78.1	60.6	75.0	59.3	65.4	78.4	62.8	74.8	60.8	71.9	6.4	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.3	94.2	73.3	58.0	86.6	69.2	56.2	80.9	67.1	33.2	78.4	31.0	75.0	29.5	72.0	73.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
23.1	20.6	17.8	-27.7	88.1	5.7	4.0	-31.8	91.0	-35.2	93.3	-38.4	95.6	-42.5	98.4		
			-27.8	68.5	5.7	3.6	-31.9	71.2	-35.2	73.3	-38.4	75.3	-42.5	77.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	33.2	-1.1	6.8	21.6	37.6	48.4	58.2	62.3	58.6	48.3	33.8	17.4	5.1	(6)
(7)		DBStd	23.61	10.88	11.11	11.72	9.28	8.55	6.45	5.17	5.56	8.10	9.51	11.76	10.78	(7)
(8)		HDD50	7197	1584	1209	879	379	134	11	0	5	122	503	978	1393	(8)
(9)		HDD65	11655	2049	1629	1344	822	517	217	113	209	502	967	1428	1858	(9)
(10)		CDD50	1075	0	0	0	8	83	257	383	272	71	1	0	0	(10)
(11)		CDD65	57	0	0	0	0	2	13	31	10	1	0	0	0	(11)
(12)		CDH74	909	0	0	0	0	60	264	393	170	22	0	0	0	(12)
(13)		CDH80	191	0	0	0	0	7	52	94	36	2	0	0	0	(13)
(14)	Wind	WSAvg	6.3	5.1	5.6	7.2	8.3	8.2	6.3	5.2	5.4	6.1	6.5	6.0	5.6	(14)
(15)	Precipitation	PrecAvg	11.1	0.1	0.1	0.1	0.4	0.8	2.2	3.1	2.8	0.9	0.3	0.1	0.1	(15)
(16)		PrecMax	15.6	0.3	0.3	0.4	1.3	2.3	4.6	8.0	7.2	2.0	1.5	0.5	0.6	(16)
(17)		PrecMin	7.0	0.0	0.0	0.0	0.0	0.0	0.7	0.7	0.4	0.1	0.0	0.0	0.0	(17)
(18)		PrecStd	2.6	0.1	0.1	0.1	0.4	0.6	1.1	1.4	1.5	0.5	0.3	0.1	0.1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.4	39.4	55.9	69.1	80.3	85.6	87.9	84.4	77.0	63.0	49.1	33.9	(19)
(20)			MCWB	24.5	31.7	41.6	49.1	56.5	64.3	67.1	66.2	55.9	48.0	38.5	27.6	(20)
(21)		2%	DB	24.0	33.4	49.4	62.9	74.6	80.5	82.5	78.7	71.5	57.9	43.0	27.9	(21)
(22)			MCWB	19.8	26.9	37.9	47.0	54.2	61.7	63.9	62.1	55.3	45.5	34.7	22.9	(22)
(23)		5%	DB	19.3	28.3	44.2	58.3	70.2	76.7	78.2	74.5	67.1	53.4	38.0	23.9	(23)
(24)			MCWB	15.9	23.3	34.5	44.0	52.3	59.3	62.0	60.2	53.5	42.3	31.3	19.7	(24)
(25)		10%	DB	14.3	23.5	39.2	54.1	65.7	73.2	74.7	70.8	63.0	48.9	33.4	20.2	(25)
(26)			MCWB	11.5	19.4	31.1	41.6	49.6	57.1	60.7	58.6	51.2	39.7	28.0	16.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.1	32.2	43.4	52.1	60.7	66.5	70.5	67.9	59.6	49.5	38.8	27.8	(27)
(28)			MCD B	29.6	38.8	54.2	66.1	76.5	81.8	85.6	82.9	73.0	61.1	47.9	33.5	(28)
(29)		2%	WB	19.9	27.2	38.6	47.8	56.9	63.0	66.4	63.5	56.5	46.0	34.9	23.1	(29)
(30)			MCD B	24.0	33.0	48.2	60.4	71.3	77.2	78.8	76.1	68.7	56.9	42.2	27.8	(30)
(31)		5%	WB	15.9	23.3	34.8	44.9	53.3	60.6	63.7	61.2	54.1	42.9	31.4	19.8	(31)
(32)			MCD B	19.2	28.3	43.7	57.3	67.6	74.3	75.1	72.0	65.6	52.6	37.6	23.7	(32)
(33)		10%	WB	11.6	19.4	31.4	42.1	50.4	58.5	61.6	59.3	51.5	39.8	28.0	16.7	(33)
(34)			MCD B	14.2	23.4	39.1	53.2	63.5	70.7	72.0	68.8	62.4	48.3	33.2	20.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	18.3	22.6	23.7	22.5	23.7	23.0	20.6	20.9	23.1	21.7	19.2	17.6	(35)
(36)			MCDBR	22.2	27.3	27.7	28.6	31.0	30.2	28.0	27.8	28.7	27.9	23.4	20.4	(36)
(37)			MCWBR	19.1	22.1	19.2	17.6	17.2	15.5	13.5	14.7	16.4	18.2	17.3	16.6	(37)
(38)		5% WB	MCDBR	22.1	26.8	27.1	27.7	29.0	27.4	24.9	24.7	27.3	27.2	22.8	20.1	(38)
(39)			MCWBR	19.1	22.0	19.6	18.5	18.2	15.8	13.7	14.0	16.5	18.5	17.1	16.7	(39)
(40)		Clear-Sky Solar Irradiance	taub	0.232	0.256	0.300	0.341	0.357	0.361	0.351	0.332	0.297	0.276	0.254	0.234	(40)
(41)	taud		2.368	2.330	2.276	2.260	2.259	2.351	2.430	2.455	2.492	2.468	2.390	2.340	(41)	
(42)	Ebn at Noon		275	290	290	288	286	285	286	288	290	279	261	256	(42)	
(43)	Edh at Noon		21	28	35	39	41	38	35	32	28	24	20	19	(43)	
(44)	All-Sky Solar Radiation	RadAvg	560	906	1387	1770	2036	2035	1937	1698	1447	972	595	449	(44)	
(45)		RadStd	22	29	48	84	121	100	114	69	60	50	28	20	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	-4.05	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	N/A	N/A	-3.64	N/A	-1.08	-1.33	N/A	N/A	N/A	N/A

Nomenclature: See separate page